

Pranjal Raj

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EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Electronics and Communication (CGPA: 8.30)

Vellore, Tamil Nadu

Aug 2024 - 2026

Relevant Coursework: Operating Systems, Database Management Systems, Computer Networks, Data Structures & Algorithms, Software Engineering, Microprocessors & Interfacing

Mount Litera Zee School

Grade XII, P.C.M with Computer Science (Perc.: 85%)

Roorkee, Uttarakhand

July 2022 - May 2024

Mount Litera Zee School

Grade X (Perc.: 96%)

Roorkee, Uttarakhand

July 2022

EXPERIENCE

Mobile Systems Engineering Intern

ASD , Kolkata

Remote

June 2026 - Present

- Built high-performance Flutter mobile app with real-time data synchronization, achieving 60 FPS and <500ms sync latency through asynchronous APIs, SQLite caching, and race condition prevention .
- Optimized concurrent state management using Dart isolates for background processing, reducing memory footprint by 40% and maintaining 99.8% app stability across test deployments.
- Implemented graceful degradation patterns for offline-first architecture, ensuring zero data loss during network transitions and reliable synchronization with cloud backend.

Full-Stack Software Engineering Intern

Kinsuk (Early-stage Startup)

On-site

April - May 2026

- Architected end-to-end full-stack application from ground-up, designing system architecture for scalability across frontend (React.js), backend (Node.js/Express.js), and database layers (MongoDB), enabling production deployment.
- Designed and implemented RESTful API with proper authentication (JWT), session management, error handling, and database indexing strategies to optimize query performance for large-scale data operations.
- Built reactive frontend system using React.js with state management (Redux concepts), implementing efficient data flow and minimizing unnecessary re-renders through component optimization techniques.
- Documented system architecture and technical design decisions; presented detailed architecture diagrams to stakeholders, facilitating successful grant acquisition and enabling external team onboarding and system maintenance.
- Implemented deployment pipeline and configured production environment, managing infrastructure concerns like database replication and application monitoring for reliability.

PROJECTS

Panchayat (Society Mobile Application)

React 18+, Typescript, Tailwind CSS, Lucid React, Vite, NodeJS, ExpressJS, Secure API Layer

June 2026 - August 2026

- Architected and built a full-stack cooperative housing management platform using React 18+, TypeScript, Tailwind CSS, and Node.js/Express.js, centralizing administrative, operational, and communication workflows for administrators, caretakers, and residents into a unified digital ecosystem.

- Implemented role-based authentication, secure profile management, and integrated subscription/payment flows, directly addressing chronic inefficiencies in record-keeping, transparency, and fee collection for housing societies.
- Designed a mobile-first, responsive UI with Tailwind CSS and Framer Motion micro-animations, delivering a polished, high-contrast user experience across all device types.
- Engineered a secure server-side API layer with Express.js to handle authentication, data management, and critical operations, ensuring sensitive data remains protected and never exposed client-side.

GitHub: [Project Link](#)

IntelliBin (Software + Hardware)

Google Firebase, HTML5, CSS, Javascript, NodeJS, Real-time Web Dashboard

March 2026 - May 2026

- Designed complete IoT system architecture for intelligent waste management, integrating hardware sensors (ultrasonic, gas, temperature, moisture, GPS) with cloud infrastructure for real-time monitoring and optimization.
- Built embedded system firmware on ESP32 using Arduino IDE with asynchronous sensor data collection, handling concurrent I/O operations and synchronization of multi-sensor inputs at real-time intervals.
- Implemented data aggregation pipeline using HTTP APIs to transmit sensor data to Firebase with custom JSON payload optimization, reducing bandwidth usage by 45% through compression and selective field transmission.
- Developed real-time web dashboard using Firebase Realtime Database with live data visualization, implementing Pub/Sub architecture for efficient event-driven updates across multiple connected devices and monitoring stations.
- Engineered intelligent alert system with threshold-based triggers for hazardous gas detection and bin overflow prediction, optimizing collection routes based on spatial data for 25% reduction in collection time.

GitHub: [Project Link](#)

TECHNICAL SKILLS

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), Computer Networks, Database Management Systems (DBMS), Operating System, Software Architecture

Languages and Frameworks: Java, Python, C/C++, JavaScript (ES6+), HTML/CSS, React, Bootstrap, Express.js, Node.js, MongoDB

Frontend & Web Technologies: React.js, HTML5, CSS3, Responsive Design, Tailwind CSS, Component Optimization

Developer Tools & Workflows: Git, VS Code, Eclipse, Chrome DevTools, Debugging Tools, Performance Profiling, System Monitoring

Performance Optimization: Web Performance Tuning, Code Optimization, Query Optimization, Caching Strategies (Redis), Real-time Dashboards

Backend & Infrastructure: Node.js, Express.js, RESTful APIs, WebSockets, Firebase (Realtime DB), Kubernetes Microservices

Database Design: MongoDB, Schema Design, Distributed Systems, Embedded Systems (ESP32, Arduino IDE), Sensor Integration

Data Analysis & Visualization: pandas, NumPy, Matplotlib, Scikit-Learn, NLTK, Real-time Dashboards

CO-CURRICULARS

Clubs and Chapter Participation

Jul. 2024 - Present

VIT, Vellore, Tamil Nadu

- As Senior Core Member at IET-VIT Student Chapter (Apr 2024 – Feb 2025), contributed to the Editorial Domain, drafted sponsorship communications, and wrote landing page content for the HackOIT V4.0 website.
- As a Senior Core Member of Juvenile Care, VIT (Jan 2025 – Present), active in the Editorial and PR-Media domains.

CERTIFICATIONS

DSA with Java (Delta Batch) – *Apna College*

MERN Fullstack Program – *Apna College*

Amazon Web Services Program – *Udemy*

Software Engineering Job Simulation – *Commonwealth Bank*